HAT-P-36b

R-0921

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-36_230512` · created 2026-07-29T08:48:10+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460076.8318 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.126 +/- 0.013
Transit depth (Rp/Rs)^2	1.58 +/- 0.32 [%]
Orbital Inclination (inc)	83.84 +/- 2.41
Airmass coefficient 1 (a1)	1.26 +/- 0.014
Airmass coefficient 2 (a2)	-0.1652 +/- 0.005
Scatter in the residuals of the lightcurve fit is	1.06 %
Best Comparison Star	None
Optimal Aperture	3.79
Optimal Annulus	16.2
Transit Duration (day)	0.087 +/- 0.009

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.126 ± 0.013 vs 0.126 (deviation 0.0σ)
Duration fit vs published	0.087 d vs 0.0925 d (deviation 0.49σ)
Mid-time O–C	-2.8 min
Depth SNR	7.8
Residual scatter	1.06 %

Light curve

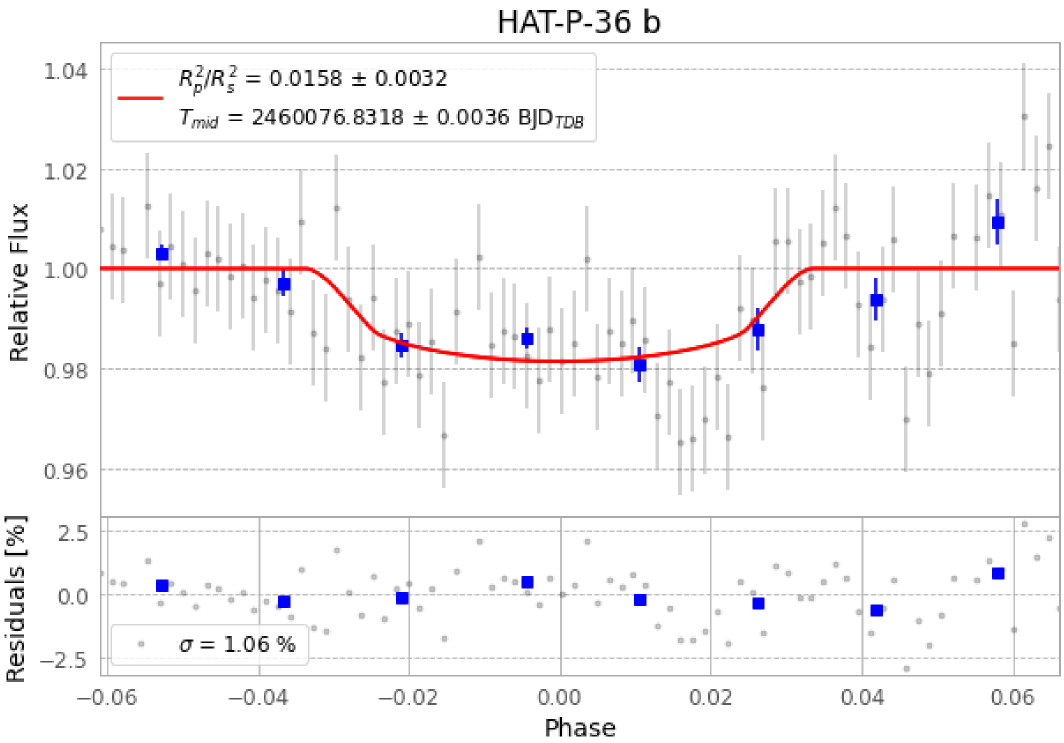

Figure 1 — FinalLightCurve_HAT-P-36 b_2023-05-11.png (SHA-256 d63cea8a80feff2b…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [295, 410], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 731c30594ab28feb…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

82 FITS frames (54 MB), `HATP-36230512055717.FITS` → `HATP-36230512100015.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-230512.log (b1fe1105b36f…), FinalLightCurve_HAT-P-36 b_2023-05-11.png (d63cea8a80fe…), FinalLightCurve_HAT-P-36 b_2023-05-11.csv (fa0cb61ae7a0…)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)